Project Title: Remote Sensing-Based Crop Mapping and Agricultural Decision Support System for Major Crops in Andhra Pradesh

Project Batch No.: 21CS15

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Academic Year: 2026 – 27

Abstract

Agriculture is an important part of Andhra Pradesh's economy. Still, effective crop monitoring and decision-making can be difficult due to limited field-level information and changing environmental and market conditions. This project aims to develop a system to map and monitor major crops and support agricultural decision-making. The study will focus on chilli, coconut, paddy, and cotton in selected agricultural regions of Guntur, Krishna, East Godavari, and West Godavari districts of Andhra Pradesh. Sentinel-2 satellite imagery will be processed using Google Earth Engine, while Google Earth Pro and QGIS will be used for study-area selection, spatial data preparation, and visualisation. Training and validation polygons will be collected to classify crops using machine learning algorithms. Vegetation indices such as NDVI will be used to analyse crop growth and vegetation health over different periods. Environmental factors such as temperature, soil conditions, and water availability will be considered to study their influence on crop growth. Historical crop information and market factors such as crop prices and demand will also be analysed. The expected outcome is a crop distribution map, estimation of cultivated areas, crop health analysis, and a decision-support system that combines agricultural, environmental, and market information. The system can help farmers, researchers, and agricultural planners in crop monitoring, resource management, and making better-informed crop-related decisions.

Problem Statement

Farmers and agricultural planners often lack accurate and timely information about crop distribution, crop growth, environmental conditions, and market trends. This makes effective crop monitoring, planning, and informed agricultural decision-making difficult.

Keywords

Remote Sensing, Sentinel-2, Google Earth Engine, Machine Learning, Crop Mapping, NDVI, Agricultural Decision Support

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